

Strain in x-direction at point (x, y)

$$\lim_{\Delta x \rightarrow 0} \frac{[PQ] - [P^0Q^0]}{[PQ]} = \lim_{\Delta x \rightarrow 0} \frac{\Delta u - \Delta x \left(1 - \frac{\partial u}{\partial x}\right)}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\epsilon_{xx} = \frac{\partial u}{\partial x}$$

Can be proved that

$$\epsilon_{yy} = \frac{\partial v}{\partial y}$$

Assignment - ① Prove that $\epsilon_{yy} = \frac{\partial v}{\partial y}$
With Complete Steps.

Last Date: 20th April.

Late Submission will be evaluated at 50%

Some Important

$$\epsilon = \text{Sym}(\nabla \vec{u}) = \frac{1}{2} \left[\nabla \vec{u} + (\nabla \vec{u})^T \right]$$

$$\forall \nabla \vec{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}$$

② $\rightarrow$ Displacement $\rightarrow \epsilon \rightarrow 2^{\text{nd}}$ Order Tensor $\checkmark$

Assignment-2: Prove this.
Read Text Book
Zimmerman

$$\epsilon' = L \epsilon L^T$$